Supplementary Material

Predicting Family Educational Climate from Naturalistic Audio Sensing of Parent-Child Homework Interactions

S1. Measure overview

The main paper modeled nine questionnaire dimensions separately. Table S1 summarizes only the measures and dimensions used in the final family-level prediction dataset.

Table S1. Questionnaire measures corresponding to the nine modeled outcomes.

Instrument	Dimension	Scored items	α	Response scale	Source
s-EMBU	Emotional warmth		.79	5-point scale: 从不 — Never 较少 — Rarely 一般 — Sometimes 经常 — Often 非常频繁 — Very often	Arrindell et al. (1999); Jiang et al. (2010)
	Overprotection		.73		
	Rejection		.84		
PERS-C	Emotional attention		.82	5-point scale: 不同意 — Disagree 比较不同意 — Somewhat disagree 中立 — Neutral 同意 — Agree 非常同意 — Strongly agree	Pereira et al. (2017); Lu et al. (2022)
	Emotional avoidance		.65		
	Emotional acceptance		.49		
	Emotional control		.65		
Parent-child communication quality	Relationship-oriented communication		.86	5-point scale: 不同意 — Disagree 比较不同意 — Somewhat disagree 中立 — Neutral 同意 — Agree 非常同意 — Strongly agree	Chi (2011)
	Problem-oriented communication		.83		

Note. s-EMBU = short-form Egna Minnen av Barndoms Uppfostran; PERS-C = Chinese version of the Parent Emotion Regulation Scale. Chinese was the language of administration; English translations of response options are provided for readability. Item counts refer to the scored items used for the modeled dimensions. Internal consistency coefficients were calculated using the available questionnaire responses in the current sample.

S2. Representative administered items

Table S2 provides one representative administered item for each modeled dimension.

Table S2. Representative Chinese items with English translations.

Instrument	Dimension	Representative administered item (Chinese and English translation)
s-EMBU	Emotional warmth	我会夸奖我的孩子。 I praise my child.
s-EMBU	Overprotection	我会干涉孩子所做的一切。 I interfere with everything my child does.
s-EMBU	Rejection	即使是小错，我也会严惩孩子。 I punish my child severely even for minor mistakes.
PERS-C	Emotional attention	我关注我孩子的情绪并试着去理解他们。 I pay attention to my child's emotions and try to understand them.
PERS-C	Emotional avoidance	如果有可能，我会消除孩子的所有负面情绪。 If possible, I would eliminate all of my child's negative emotions.
PERS-C	Emotional acceptance	当我的孩子紧张时，我不是立即采取行动，而是给他时间让他自己冷静下来并解决问题。 When my child is distressed, I do not act immediately; instead, I give them time to calm down and solve the problem on their own.
PERS-C	Emotional control	当我的孩子紧张时，我能够保持冷静并帮助他/她面对这种情况。 When my child is distressed, I am able to remain calm and help them face the situation.
Parent-child communication quality	Relationship-oriented communication	孩子与我的交流让彼此更加亲密。 Communication between my child and me makes us feel closer.
Parent-child communication quality	Problem-oriented communication	孩子与我的讨论有助于解决问题。 Discussions between my child and me help solve problems.

Note. s-EMBU = short-form Egna Minnen av Barndoms Uppfostran; PERS-C = Chinese version of the Parent Emotion Regulation Scale. Chinese items are shown as administered; English translations were prepared for this supplement and were not administered to participants.

S3. Prediction uncertainty and target correspondence

Tables S3A and S3B report paired comparisons between each final audio model and its corresponding non-informative baseline using the saved out-of-fold predictions, together with Pearson correlations between out-of-fold predictions and observed questionnaire scores. Target-specific sample sizes reflect the availability of valid questionnaire scores for each outcome.

Table S3A. Out-of-fold MAE estimates and uncertainty.

Dimension and comparison	n	Audio MAE	Baseline MAE	MAE reduction (%)	95% CI for MAE difference
Relationship-oriented communication SVR-RBF vs mean baseline	62	0.348	0.368	+5.43	[-0.0201, 0.0584]
Problem-oriented communication SVR-linear vs median baseline	62	0.222	0.227	+2.20	[-0.0188, 0.0288]
Emotional warmth Random forest vs mean baseline	65	0.368	0.372	+1.08	[-0.0383, 0.0453]
Overprotection XGBoost vs median baseline	65	0.446	0.433	-3.00	[-0.0699, 0.0442]
Rejection SVR-linear vs median baseline	65	0.436	0.473	+7.82	[-0.0211, 0.0916]
Emotional attention SVR-linear vs mean baseline	62	0.379	0.420	+9.76	[-0.0107, 0.0919]
Emotional avoidance KNN vs median baseline	62	0.715	0.724	+1.24	[-0.0602, 0.0759]
Emotional acceptance HistGB vs median baseline	62	0.493	0.492	-0.20	[-0.0017, 0.0001]
Emotional control Ridge vs mean baseline	62	0.559	0.660	+15.30	[-0.0128, 0.2127]

Note. MAE difference was calculated as baseline absolute error minus audio-model absolute error; positive values favor the audio model. Percentage reductions were calculated from the three-decimal MAE values shown in the main paper. Percentile 95% confidence intervals were estimated from 50,000 paired family-level bootstrap resamples of unrounded out-of-fold absolute-error differences.

Table S3B. Pearson correlations between out-of-fold predictions and observed scores.

Dimension	n	Pearson's r	95% family-level bootstrap CI
Relationship-oriented communication	62	0.226	[0.0005, 0.4342]
Problem-oriented communication	62	0.213	[0.0145, 0.3978]
Emotional warmth	65	0.205	[-0.0239, 0.4277]
Overprotection	65	0.055	[-0.1789, 0.3088]
Rejection	65	0.339	[0.1353, 0.5419]
Emotional attention	62	0.229	[-0.0137, 0.4404]
Emotional avoidance	62	0.105	[-0.1833, 0.3824]
Emotional acceptance	62	-0.227	[-0.4789, 0.0522]
Emotional control	62	0.474	[0.2614, 0.6520]

Note. Pearson's r was calculated from the saved out-of-fold predictions; percentile 95% confidence intervals were estimated from 50,000 family-level bootstrap resamples.

S4. Copyright and reuse note

The questionnaire instruments and their Chinese adaptations originate from the cited publications and may be subject to copyright or other reuse restrictions. This supplement therefore reports measure structure and a limited number of representative items for scholarly transparency, rather than redistributing complete instruments. Researchers seeking to reuse complete questionnaires should consult the original sources and obtain any permissions required by the relevant rights holders.

References

Arrindell, W. A., Sanavio, E., Aguilar, G., Sica, C., Hatzichristou, C., Eisemann, M., Recinos, L. A., Gaszner, P., Peter, M., Battagliese, G., Kállai, J., & van der Ende, J. (1999). The development of a short form of the EMBU: Its appraisal with students in Greece, Guatemala, Hungary and Italy. *Personality and Individual Differences, 27*(4), 613–628. https://doi.org/10.1016/S0191-8869(98)00192-5

Chi, L.-P. (2011). A three-level model of parent-child communication: Theory, scale and application in Chinese 5th–6th graders. *Psychological Development and Education, 27*(2), 140–150.

Jiang, J., Lu, Z.-R., Jiang, B.-J., & Xu, Y. (2010). Revision of the short-form Egna Minnen av Barndoms Uppfostran for Chinese. *Psychological Development and Education, 26*(1), 94–99.

Lu, Y.-B., Huang, X.-J.-Q., Long, J.-J., & Wu, W.-F. (2022). The revision of the Chinese version of the Parental Emotion Regulation Scale and its testing of reliability and validity. *Chinese Journal of Clinical Psychology, 30*(3), 536–540.

Pereira, A. I., Barros, L., Roberto, M. S., & Marques, T. (2017). Development of the Parent Emotion Regulation Scale (PERS): Factor structure and psychometric qualities. *Journal of Child and Family Studies, 26*(12), 3327–3338. https://doi.org/10.1007/s10826-017-0847-9